

WEEK-6:

Task 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char input[200];
```

```
    char output[200];
```

```
    int i, j = 0;
```

```
    printf("Enter text with escape sequences: ");
```

```
    fgets(input, sizeof(input), stdin);
```

```
    // Remove newline
```

```
    input[strcspn(input, "\n")] = '\0';
```

```
    // Process escape sequences
```

```
    for (i = 0; input[i] != '\0'; i++) {
```

```
        if (input[i] == '\\' && input[i + 1] != '\0') {
```

```
            // Escaped space
```

```
            if (input[i + 1] == ' ') {
```

```
                output[j++] = ' ';
```

```
    i++;  
}
```

```
// Escaped special symbols
```

```
else if (input[i + 1] == '$' ||  
        input[i + 1] == '&' ||  
        input[i + 1] == '|' ||  
        input[i + 1] == ';' ||  
        input[i + 1] == '\\') {  
    output[j++] = input[i + 1];  
    i++;  
}
```

```
// New line escape
```

```
else if (input[i + 1] == 'n') {  
    output[j++] = '\n';  
    i++;  
}
```

```
// Tab escape
```

```
else if (input[i + 1] == 't') {  
    output[j++] = '\t';  
    i++;  
}
```

```
// Unknown escape sequence
```

```
        else {
            output[j++] = '\\';
        }
    }
    else {
        output[j++] = input[i];
    }
}

output[j] = '\\0';

printf("\n--- Parser Result ---\n");
printf("Original input : %s\n", input);
printf("Processed text : %s\n", output);

printf("\n--- Validation ---\n");

if (strlen(output) > 0) {
    printf("Parser Output : VALID\n");
    printf("Escape Processing : SUCCESS\n");
    printf("Characters Preserved : YES\n");
} else {
    printf("Parser Output : INVALID\n");
}

return 0;
```

```
}
```

```
quoted command test: FAILED
ashwithreddy_earla@AshwithReddy:~$ nano skill6.c
ashwithreddy_earla@AshwithReddy:~$ gcc skill6.c -o skill6
ashwithreddy_earla@AshwithReddy:~$ ./skill6
Enter text with escape sequences: Hello\ World

--- Parser Result ---
Original input : Hello\ World
Processed text : Hello World

--- Validation ---
Parser Output : VALID
Escape Processing : SUCCESS
Characters Preserved : YES
```

Task 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/types.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    pid_t pid;
```

```
    int status;
```

```
    printf("Parent Process Started\n");
```

```
    printf("Parent PID: %d\n", getpid());
```

```
    // Create child process
```

```
    pid = fork();
```

```
if (pid < 0) {
    perror("fork failed");
    return 1;
}

// Child process
else if (pid == 0) {

    printf("\nChild Process Created\n");
    printf("Child PID: %d\n", getpid());
    printf("Parent PID: %d\n", getppid());

    printf("Executing command: ls -l\n");

    // Execute program and pass arguments
    execlp("ls", "ls", "-l", NULL);

    // This executes only if execlp fails
    perror("Execution failed");
    exit(1);
}

// Parent process
else {

    printf("\nParent is waiting for child...\n");
```

```
// Wait for child process
waitpid(pid, &status, 0);

if (WIFEXITED(status)) {
    printf("\nChild process finished.\n");
    printf("Child exit status: %d\n",
        WEXITSTATUS(status));
} else {
    printf("\nChild process terminated abnormally.\n");
}

printf("Parent process completed.\n");
}

return 0;
}
```

```

ashwithreddy_earla@AshwithReddy:~$ nano skill6.2.c
ashwithreddy_earla@AshwithReddy:~$ gcc skill6.2.c -o skill6.2
ashwithreddy_earla@AshwithReddy:~$ ./skill6.2
Parent Process Started
Parent PID: 1178

Parent is waiting for child...

Child Process Created
Child PID: 1179
Parent PID: 1178
Executing command: ls -l
total 752
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 779 Aug 25 03:26 Experiment2.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16304 Aug 25 03:33 Experiment5
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1353 Aug 25 03:33 Experiment5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1600 Aug 31 05:25 Experiment6server.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16304 Sep 20 15:45 Experiment7.1
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 627 Sep 20 15:45 Experiment7.1.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16256 Sep 20 15:35 Experiment8
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16216 Sep 20 16:00 Experiment8.1
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1430 Sep 20 16:00 Experiment8.1.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16336 Sep 20 16:08 Experiment8.2
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1526 Sep 20 16:07 Experiment8.2.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 849 Sep 20 15:34 Experiment8.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1118 Jul 28 04:04 HELLO
-rwxr-xr-x 11 ashwithreddy_earla ashwithreddy_earla 4096 Aug 4 03:32 OSSP
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 11 03:14 OSSP.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1014 Sep 21 13:30 Skill5.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16120 Aug 23 06:02 close
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 489 Aug 23 06:01 close.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 30 Aug 25 03:22 copy.txt
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 48 Sep 10 15:51 destination.txt
-rwxr-xr-x 2 ashwithreddy_earla ashwithreddy_earla 4096 Jul 28 03:23 dirname
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 15992 Aug 11 03:43 exec
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 199 Aug 11 03:43 exec.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16392 Aug 25 03:07 exp1
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 791 Aug 25 03:04 exp1.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 99 Aug 3 05:56 exp1.program1.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16000 Jul 31 09:57 exp1.program2
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 207 Aug 3 05:59 exp1.program3.c

```

```

rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 609 Aug 11 03:49 fork_exec.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1118 Jul 28 04:07 gcc
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16216 Aug 18 03:32 input
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 510 Aug 18 03:32 input.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 18 03:36 input.txt
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 28 04:05 mkfifo
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 28 04:06 myfifo
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16120 Aug 23 05:52 open
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 443 Aug 23 05:51 open.c
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16000 Jul 31 09:38 operating
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 58 Aug 23 05:57 output.txt
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16864 Aug 24 05:22 produce
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1868 Aug 24 05:21 producure.c
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16168 Aug 23 05:54 read
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 693 Aug 23 05:54 read.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 908 Jul 31 10:14 sample.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1 Aug 3 06:09 sample.c.save
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 30 Aug 23 15:05 sample.txt
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16344 Aug 24 15:09 skill
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 774 Aug 25 04:06 skill.c
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16568 Aug 25 04:13 skill3
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 2851 Aug 25 04:12 skill3.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 25 Aug 25 04:12 skill3.c:Zone.Identifier
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 2377 Aug 25 04:12 skill4.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 25 Aug 25 04:12 skill4.c:Zone.Identifier
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16264 Sep 22 06:00 skill5
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16440 Sep 22 06:08 skill5.2
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1707 Sep 22 06:07 skill5.2.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1041 Sep 22 06:00 skill5.c
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16176 Sep 22 07:34 skill6
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16352 Sep 22 07:37 skill6.2
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1271 Sep 22 07:36 skill6.2.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1798 Sep 22 07:33 skill6.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 48 Aug 25 03:28 source.txt
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 23 05:57 ssize_t
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 27 Aug 28 03:38 text.c
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 19 Aug 28 04:05 text.c.save
rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16168 Aug 23 05:57 write
rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 637 Aug 23 05:56 write.c

```

```

hild process finished.
hild exit status: 0
arent process completed.

```